



MCA FINAL YEAR PROJECT · 2025–26

MEMOIR : Digital Memory Forensics System

Using RAG and Vector Databases

Intelligent, context-aware forensic analysis powered by Retrieval-Augmented Generation and semantic vector search

INTRODUCTION

The Digital Evidence Crisis

Digital communication has grown exponentially — emails, chats, documents, and cloud data now form the backbone of modern investigations. Yet traditional forensic methods struggle to keep pace.

Data Explosion

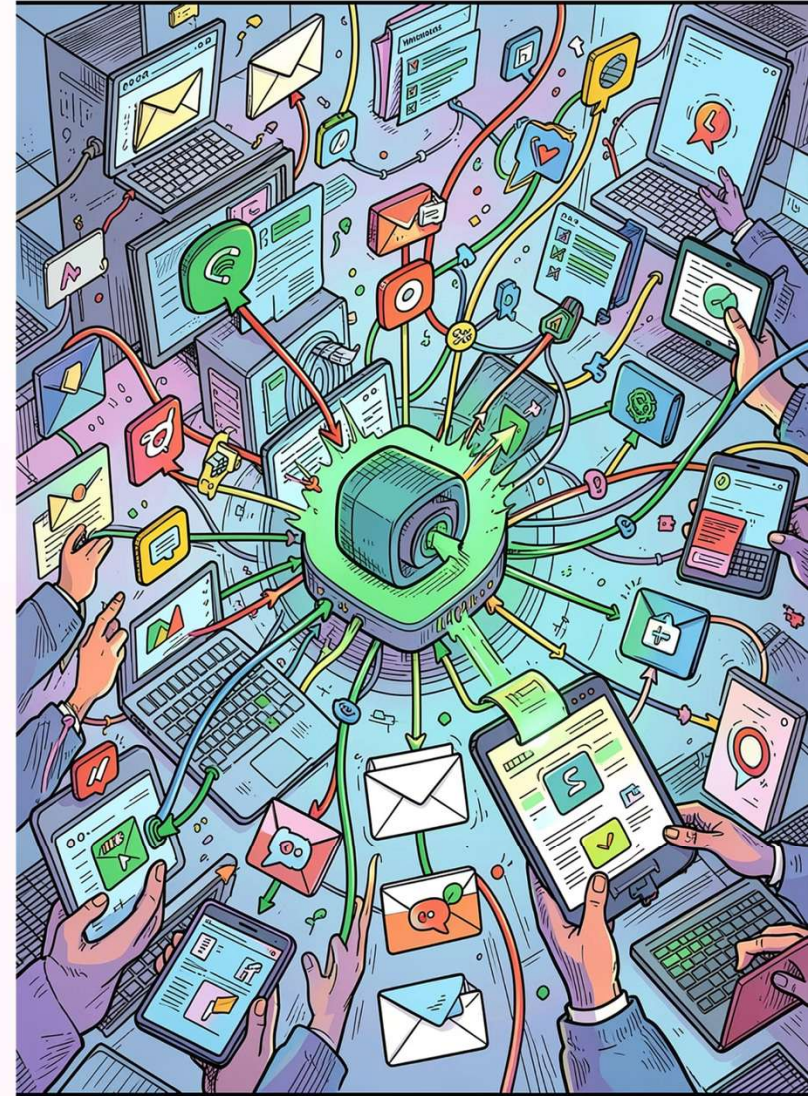
Petabytes of unstructured evidence generated daily across devices and platforms

Context Gap

Keyword searches miss semantic meaning, relationships, and hidden patterns

Time Pressure

Manual investigation is slow, error-prone, and cannot scale to modern case volumes



Existing System vs. Proposed System

Existing Approach

Manual investigation — analysts review evidence by hand

Keyword search — rigid, context-blind queries

Siloed tools — no unified intelligence layer

Slow turnaround — days to weeks per case

Proposed AI-Powered Platform

RAG Engine

Retrieval-augmented generation for contextual answers

Vector Database

Semantic similarity search across all evidence

LLM Integration

LLM models for intelligent analysis

Automated Reports

Instant, structured forensic reports

TECHNOLOGY STACK

Technologies Used



Python & Django

Core backend framework for API development and web interface



Local LLM : Ollama

Runs open-source language models locally for private, secure, and efficient AI inference.



ChromaDB

High-performance vector databases for semantic similarity search



Sentence Transformers

Embedding models converting text into dense vector representations



Cloud LLM Model : GLM5

Advanced cloud-based language model delivering powerful reasoning, generation, and contextual understanding.

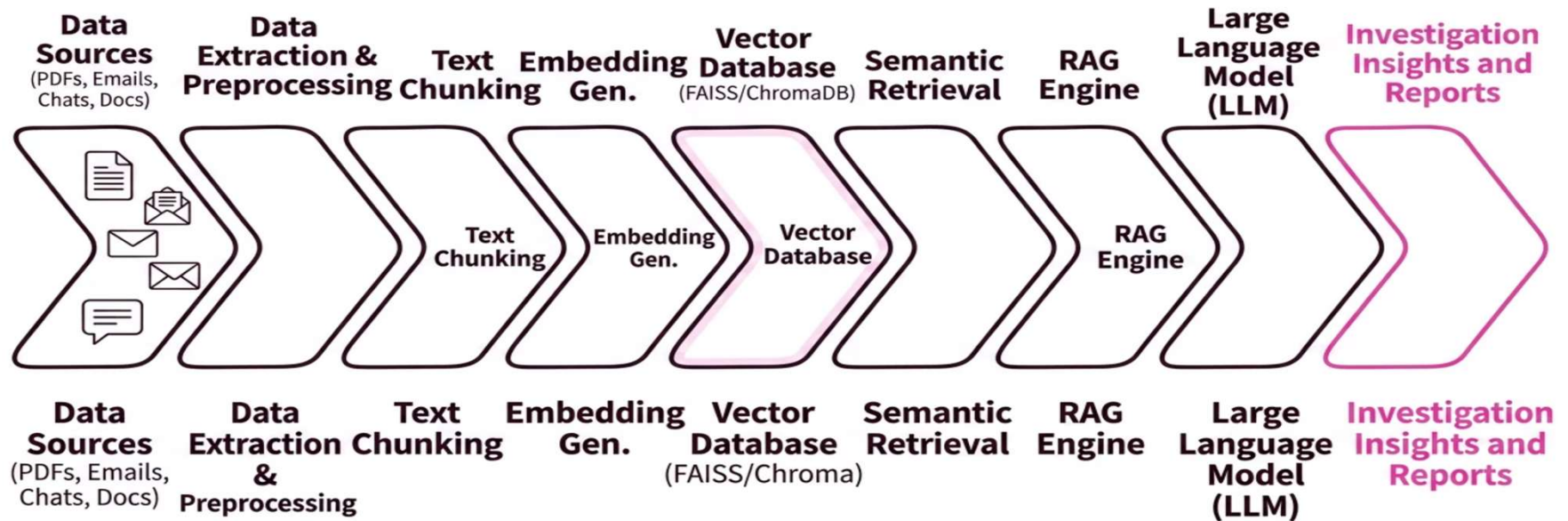


SQLite

Relational databases for structured metadata and case management

System Architecture

The end-to-end pipeline transforms raw forensic evidence into intelligent, context-aware investigative insights through a structured AI processing flow.



Each stage in the pipeline is designed for modularity — allowing individual components like the embedding model or vector store to be swapped without disrupting the overall system.

METHODOLOGY

Working Methodology

Step-by-Step Workflow

01

Data Collection

Ingest PDFs, emails, chats, and documents

02

Preprocessing

Clean, normalize, and extract text content

03

Embedding Creation

Generate dense vector representations via transformers

04

Vector Storage

Persist embeddings in ChromaDB

05

Semantic Search

Query evidence by meaning, not keywords

06

RAG Processing

Retrieve relevant chunks, augment LLM prompt

07

Analysis Results

Generate insights, timelines, and reports



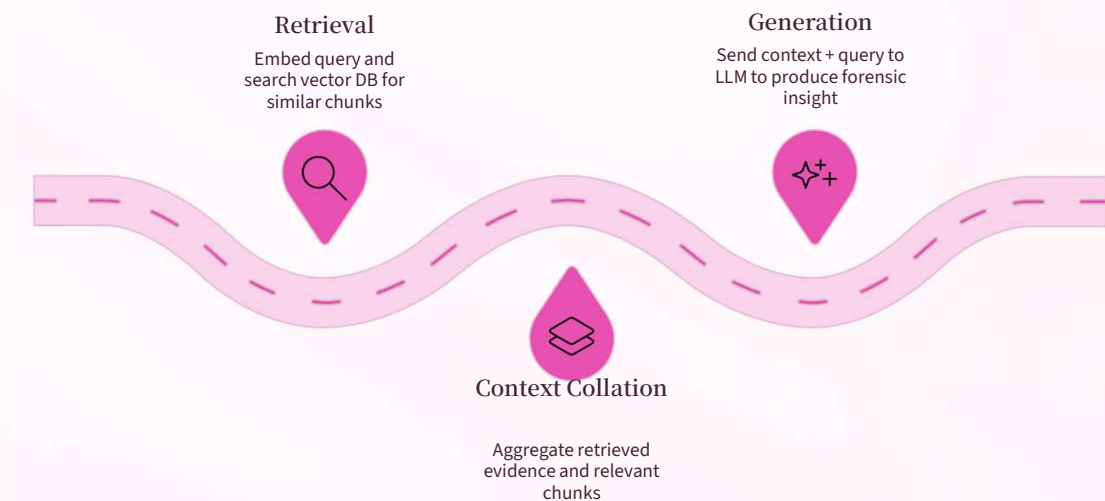
CORE CONCEPTS

Vector Databases & RAG

Vector Database Concept

Text is converted into **high-dimensional vectors** (embeddings) that capture semantic meaning. Similarity search (cosine similarity) finds related evidence without exact keyword matches.

Retrieval-Augmented Generation (RAG)



RAG combines the precision of vector retrieval with the reasoning power of LLMs — enabling the system to answer complex forensic queries grounded in actual evidence.

Embeddings — numeric representations of meaning

Similarity Search — finds semantically related content

Benefits — faster, smarter than keyword search

FEATURES

Key Features & Modules



Semantic Search

Meaning-based retrieval across all evidence types



Context Chatbot

Ask natural-language questions about case evidence



Subscription Based Models

Based On Subscription and Features related to it



Anomaly Detection

Flags unusual patterns, keywords, and access events



Auto Reports

Structured forensic reports generated instantly



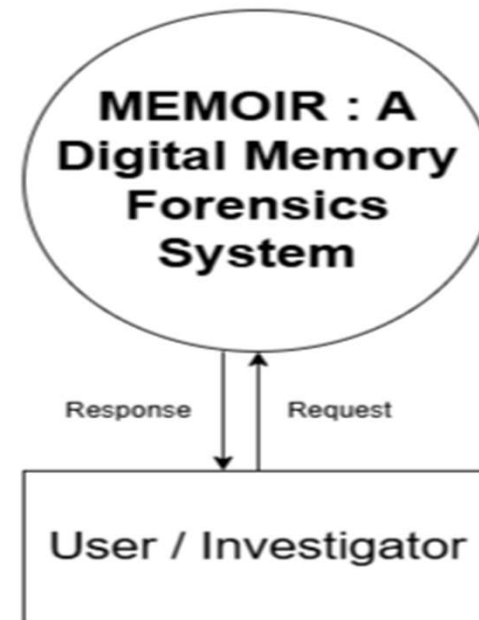
Semantic Scoring

Shows and Analysis of Semantic Scores based on Matching

FEATURES

Data Flow Diagram

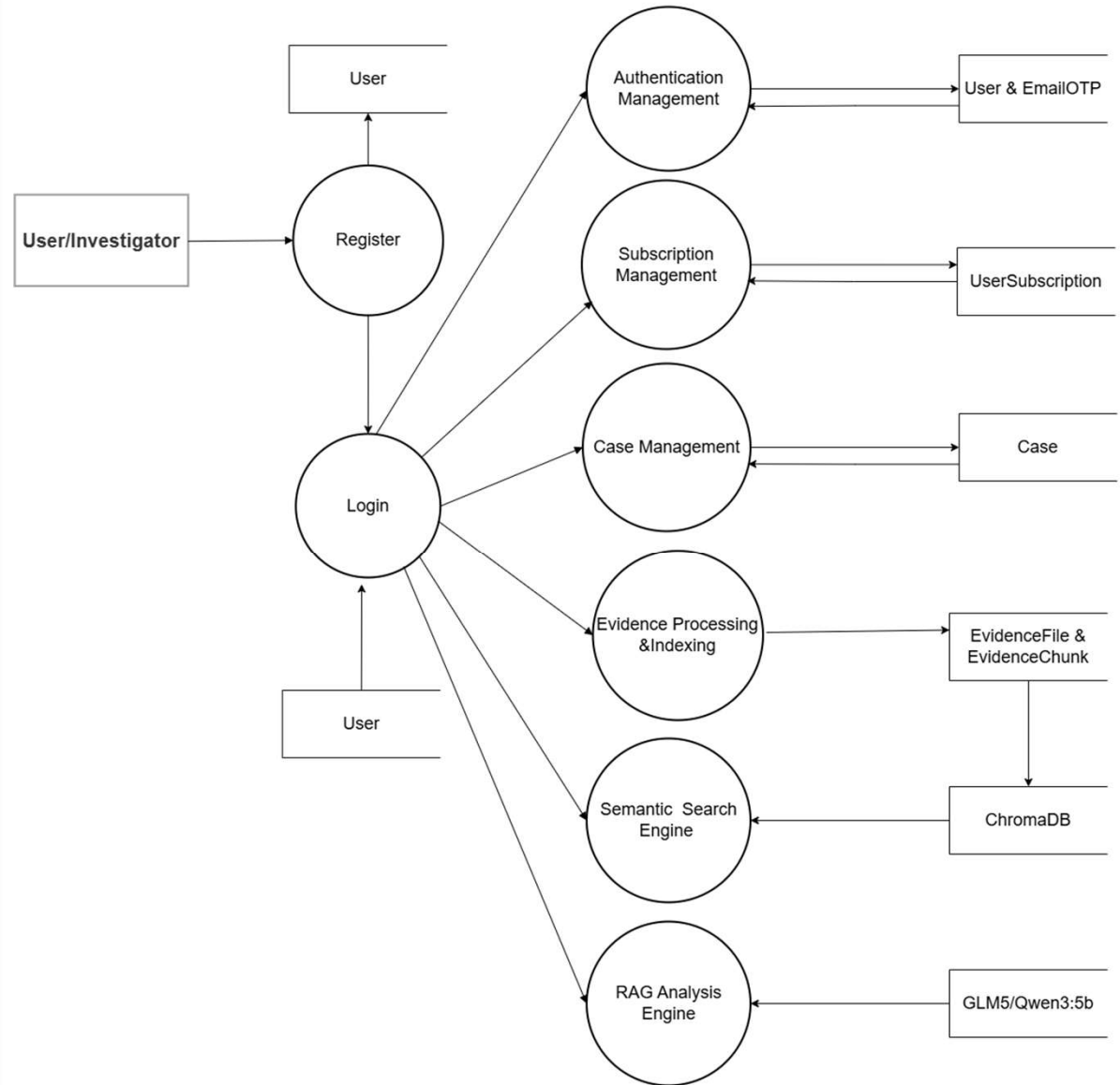
Context Level



FEATURES

Data Flow Diagram

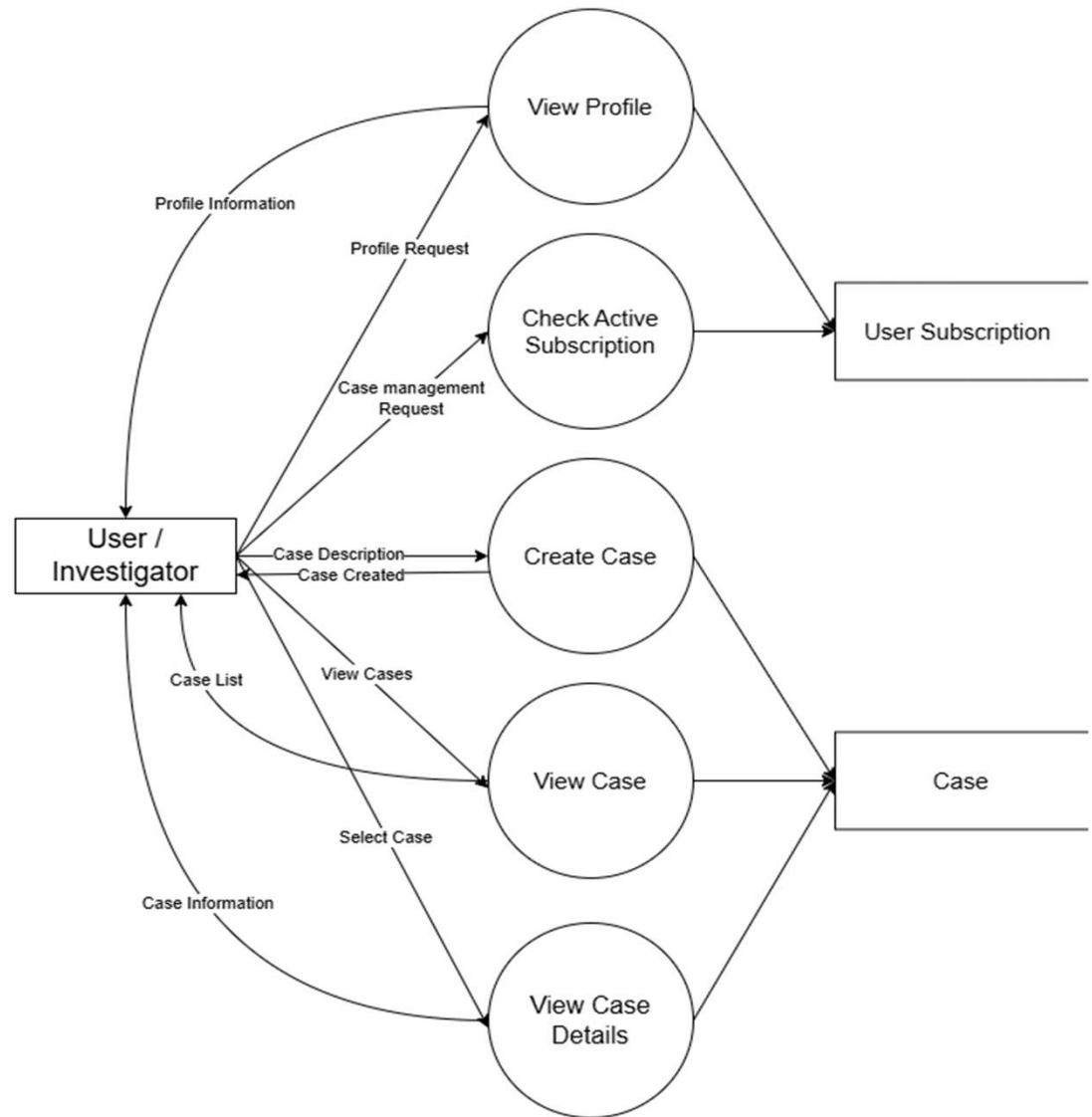
Level 1



FEATURES

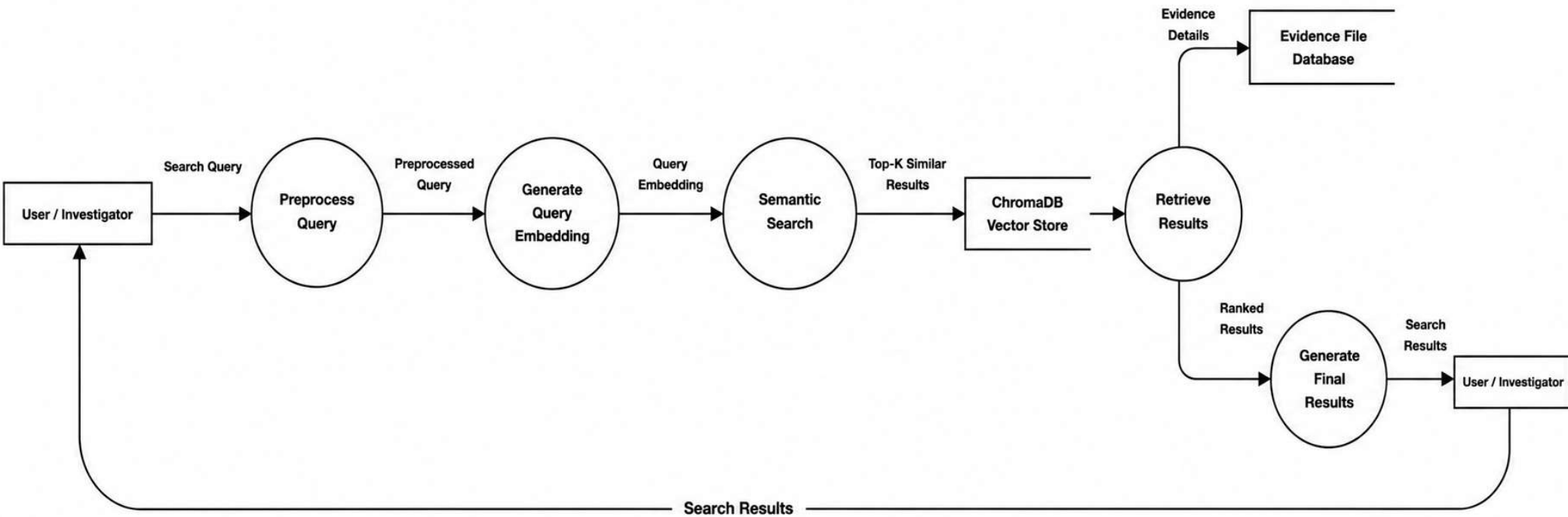
Data Flow Diagram Level 2

Cases



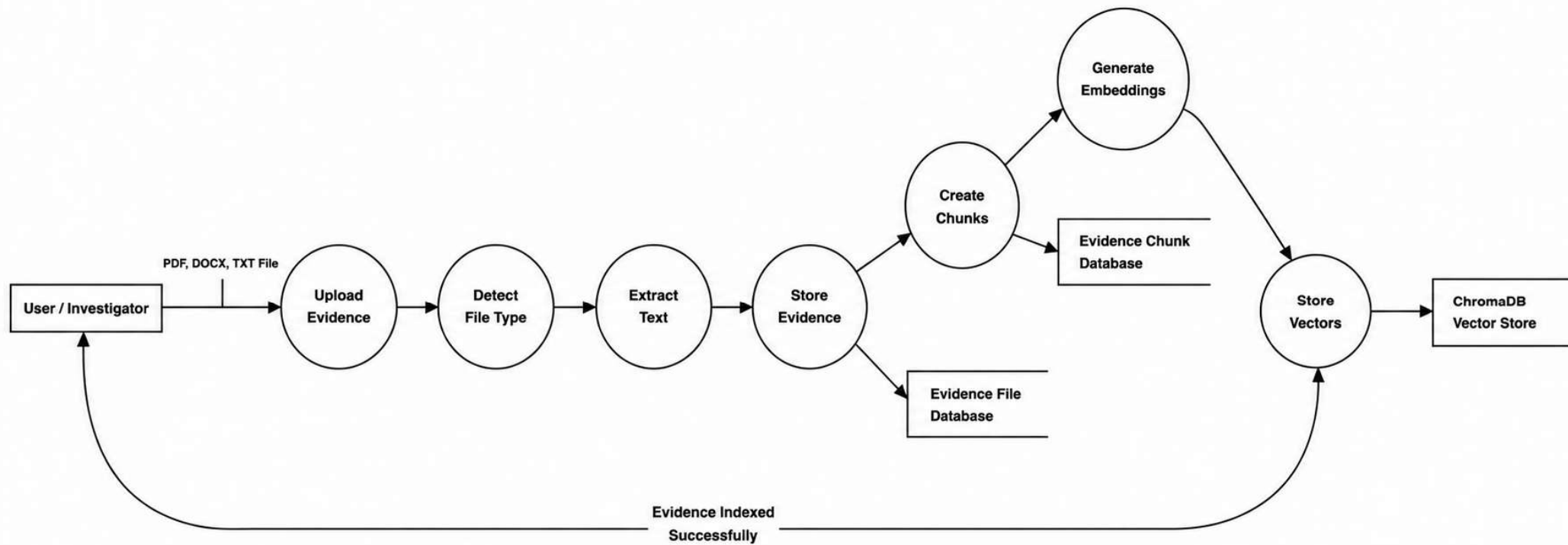
FEATURES

Data Flow Diagram- Semantic Search and Analysis



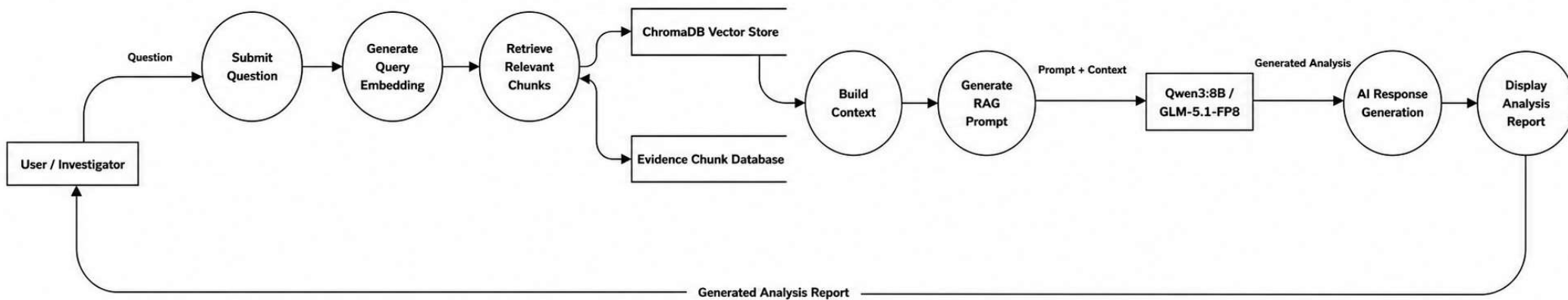
FEATURES

Data Flow Diagram - Evidence Analysis



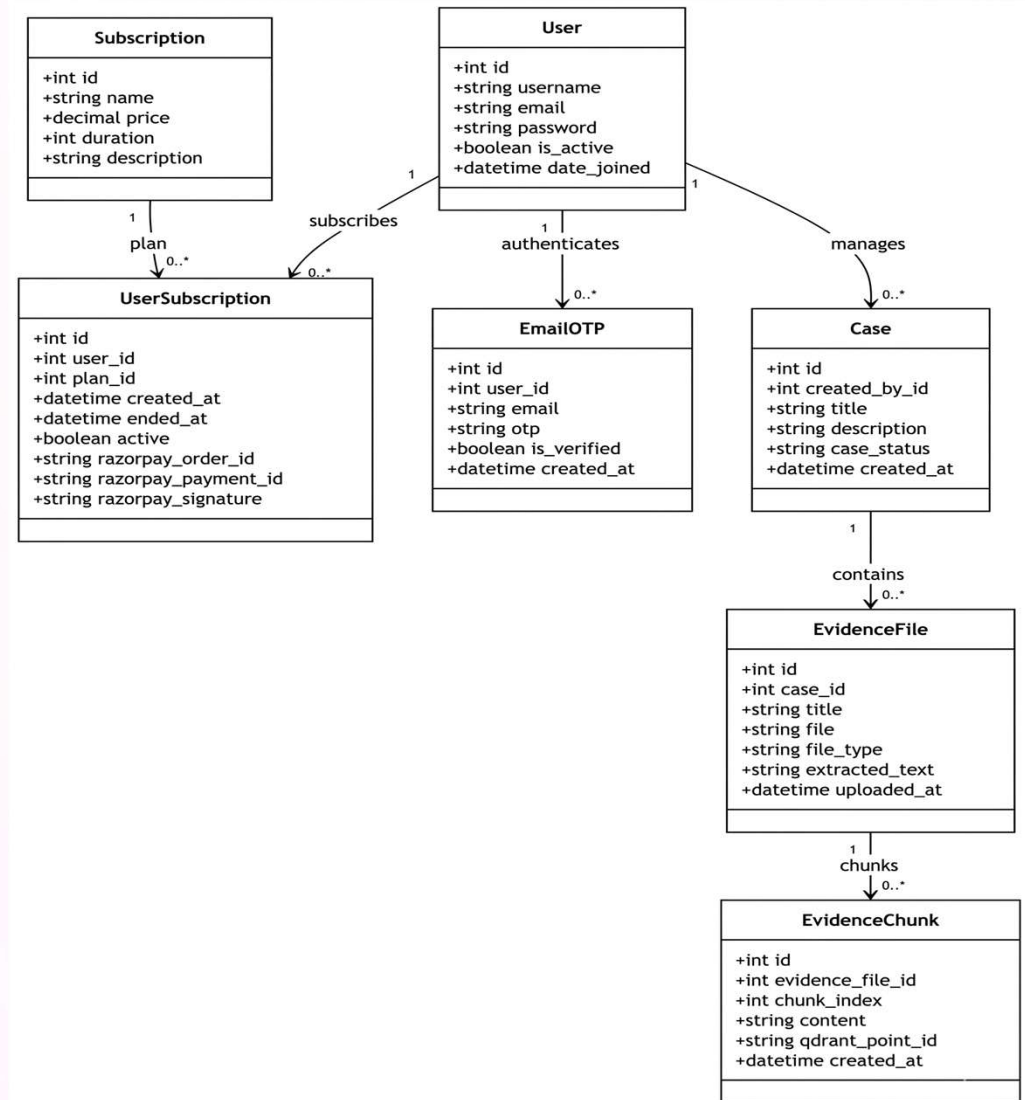
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Data Flow Diagram - RAG Analysis



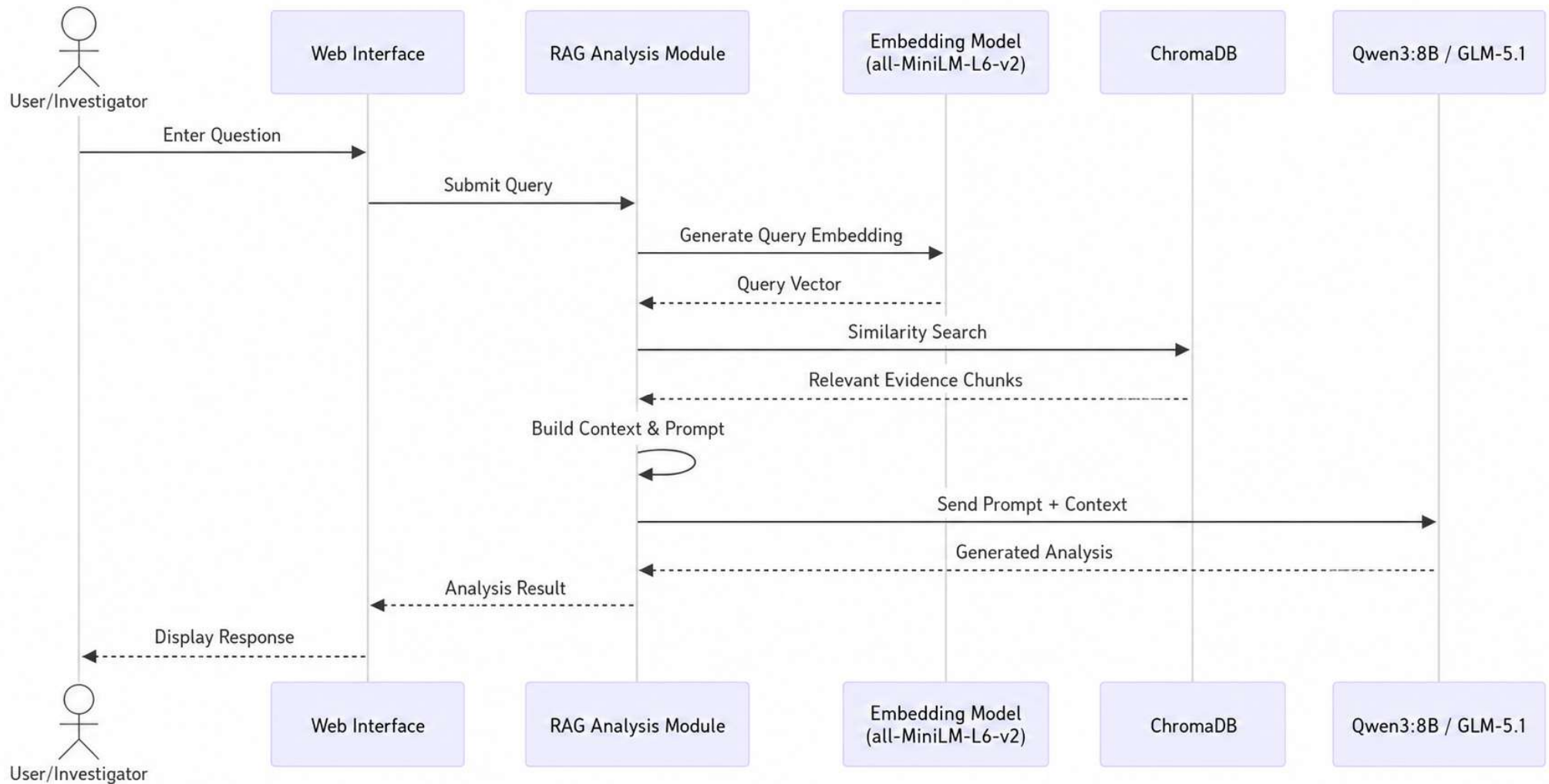
UML DIAGRAMS

Class Diagram

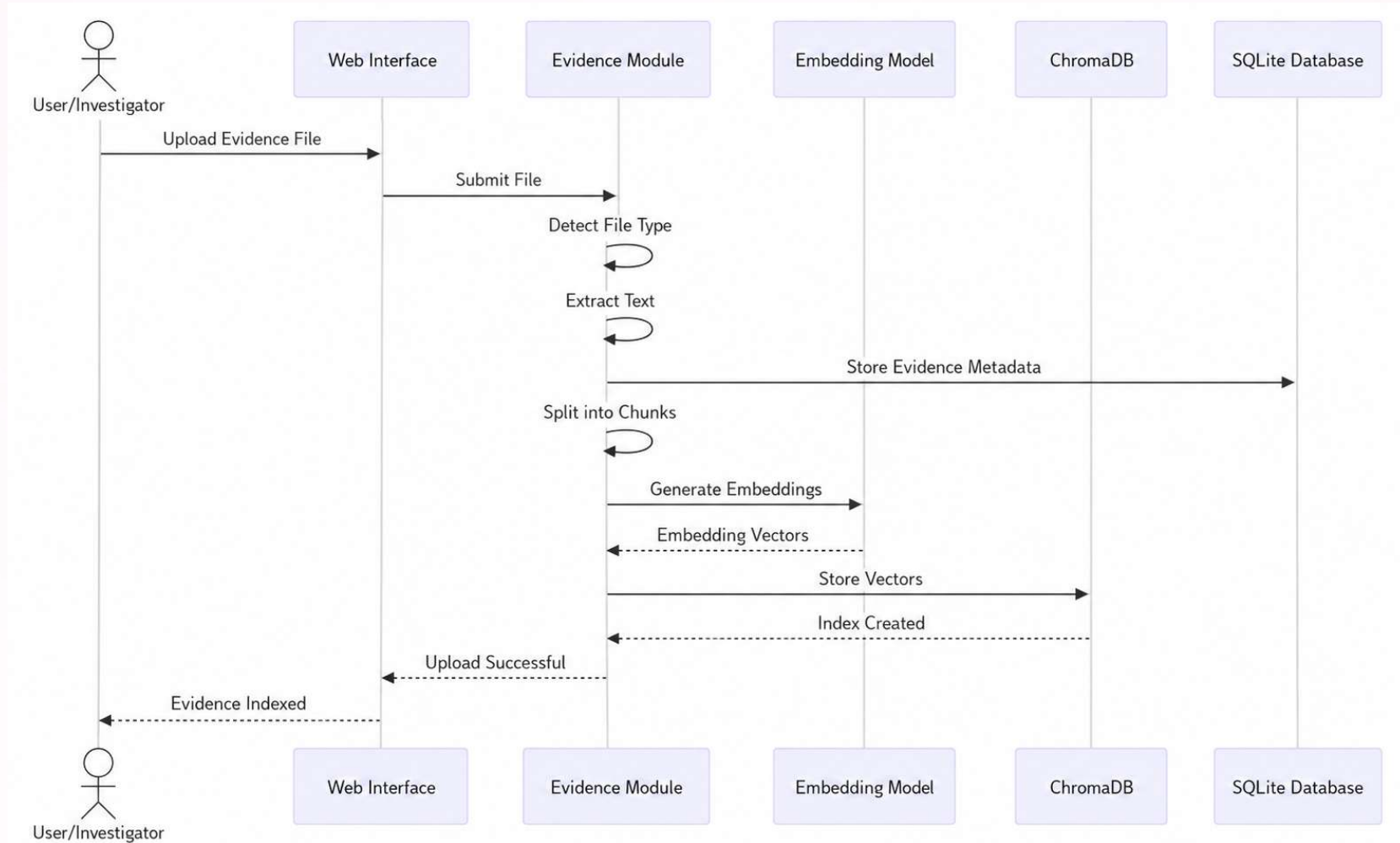


FEATURES

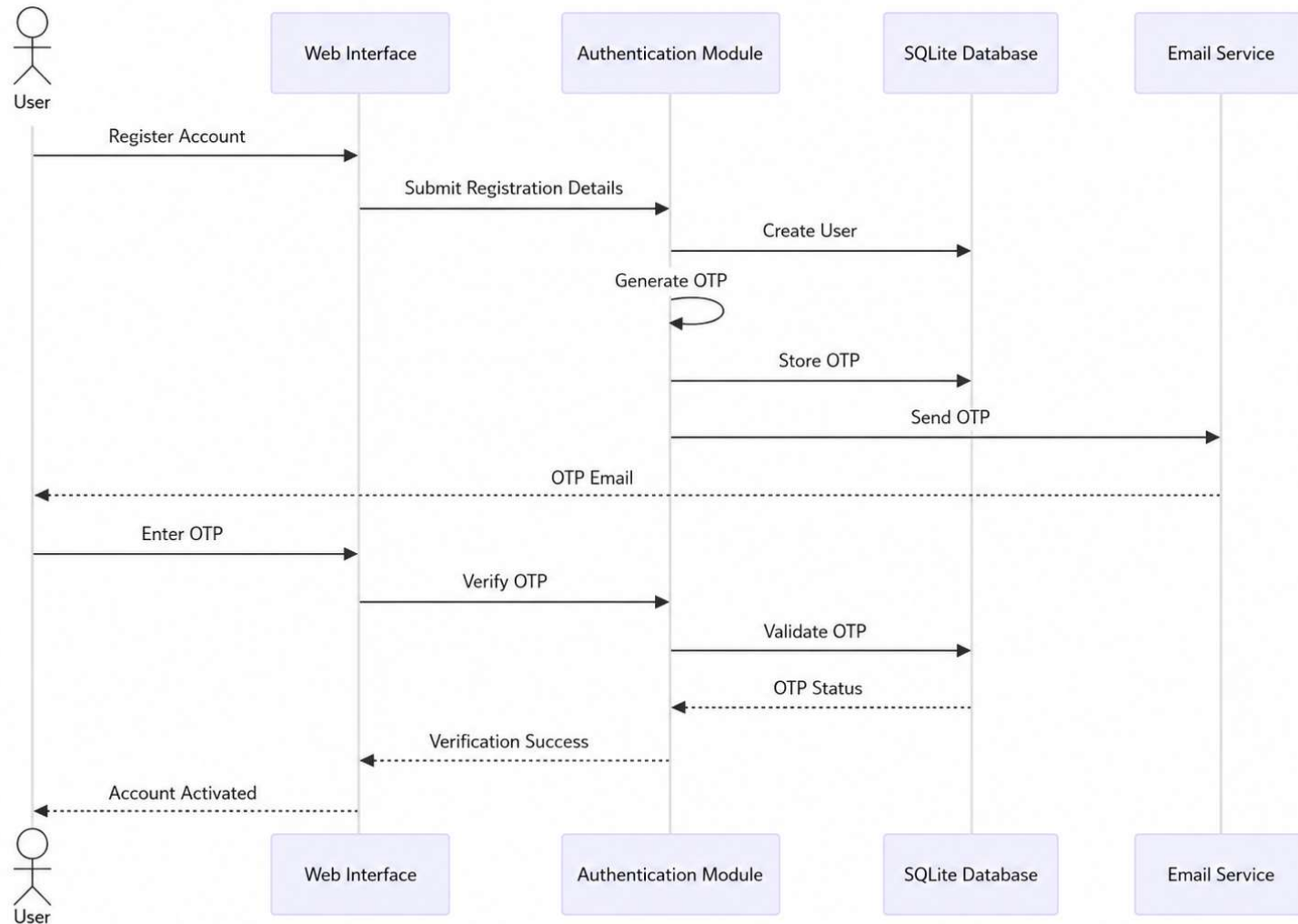
Sequence Diagram – RAG Evidence Analysis



Sequence Diagram – Evidence Upload & Indexing



Sequence Diagram – User Registration and OTP Verification



Use Case Diagram

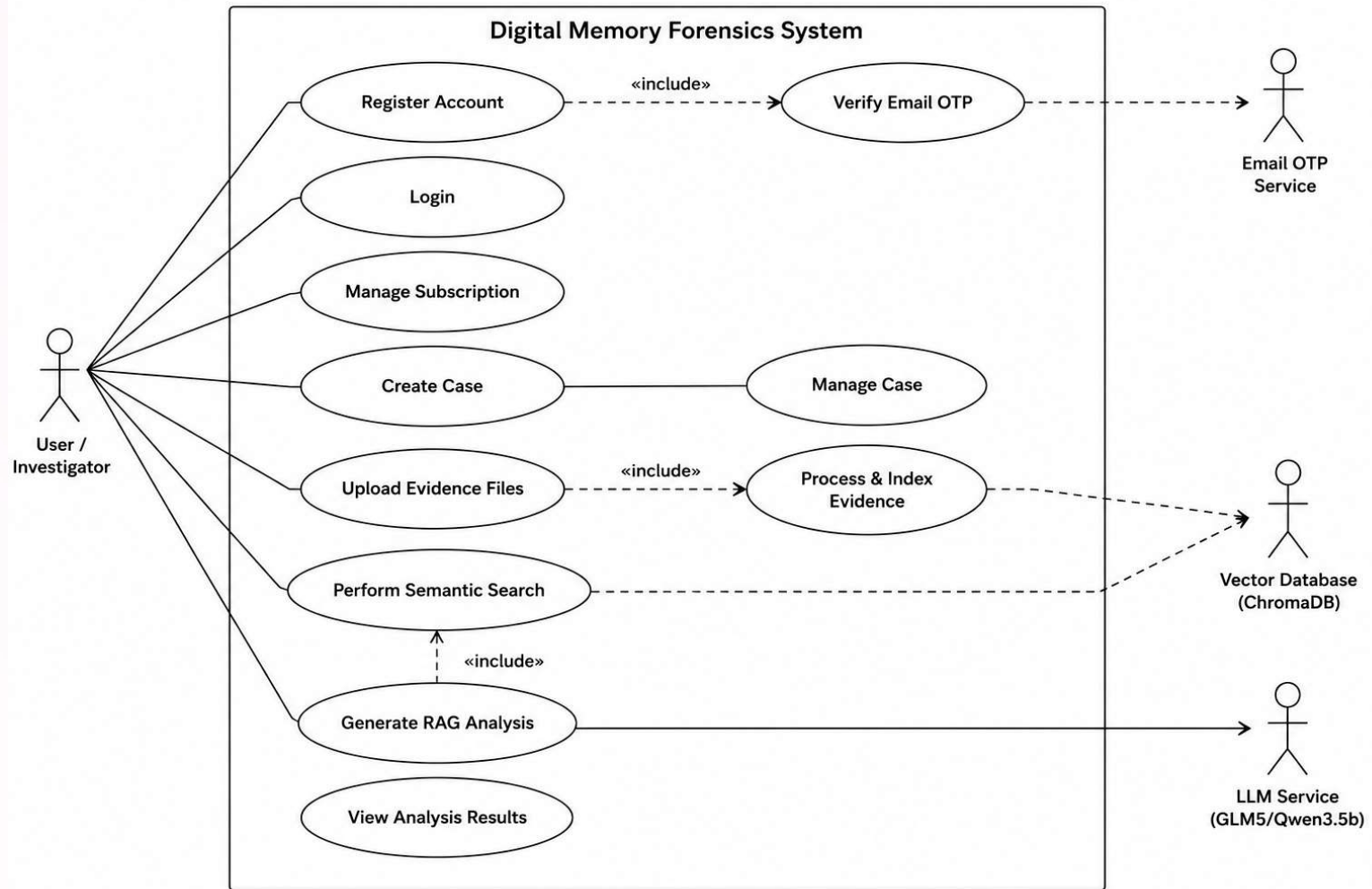


Table Structure

Column Name	Data Type	Key Type	Constraints / Description
<code>id</code>	Integer	PK	Auto-incrementing primary key
<code>username</code>	VarChar(150)	Unique	Username for logging in
<code>email</code>	VarChar(254)	-	Email address of the user
<code>password</code>	VarChar(128)	-	Cryptographically hashed password
<code>is_active</code>	Boolean	-	Flag indicating active/inactive users
<code>is_staff</code>	Boolean	-	Grants Django admin dashboard access
<code>date_joined</code>	DateTime	-	Timestamp when account was created

User

Table Structure

Column Name	Data Type	Key Type	Constraints / Description
<code>id</code>	BigInteger	PK	Auto-incrementing primary key
<code>user_id</code>	Integer	FK (OneToOne -> <code>auth_user</code>)	Direct single link to the User
<code>email</code>	VarChar(254)	-	Email address used for verification
<code>otp</code>	VarChar(6)	-	Generated 6-digit OTP verification code
<code>is_verified</code>	Boolean	-	Default: <code>False</code> . Switched to <code>True</code> upon OTP match
<code>created_at</code>	DateTime	-	Automatically set on creation

EmailOtp

Column Name	Data Type	Key Type	Constraints / Description
<code>id</code>	BigInteger	PK	Auto-incrementing primary key
<code>title</code>	VarChar(255)	-	Text title of the case folder
<code>description</code>	Text	-	Optional context/notes about the case
<code>created_by_id</code>	Integer	FK (<code>auth_user</code>)	Links to the creator (Investigator)
<code>status</code>	VarChar(20)	-	Choices: <code>open</code> (Default) or <code>closed</code>
<code>created_at</code>	DateTime	-	Automatically set on case creation

Cases

Table Structure

Column Name	Data Type	Key Type	Constraints / Description
id	BigInteger	PK	Auto-incrementing primary key
name	VarChar(100)	-	Name of the plan (e.g., SILVER, GOLD)
price	Decimal(10, 2)	-	Monetary cost of the subscription plan
duration	Integer	-	Plan duration (validity period) in days
description	Text	-	Summary details and inclusions

Subscription

Column Name	Data Type	Key Type	Constraints / Description
id	BigInteger	PK	Auto-incrementing primary key
user_id	Integer	FK (auth_user)	Link to purchasing user
plan_id	BigInteger	FK (subscription_subscription)	Link to purchased plan description
created_at	DateTime	-	Automatically set on purchase
ended_at	DateTime	-	Date/time when the subscription expires
active	Boolean	-	Default: True . Marks current active package status
razorpay_order_id	VarChar(200)	-	Razorpay unique order reference (nullable)
razorpay_payment_id	VarChar(200)	-	Razorpay transaction billing reference (nullable)
razorpay_signature	VarChar(500)	-	Cryptographic signature verifying checkout authenticity (nullable)

User Subscription

Table Structure

Column Name	Data Type	Key Type	Constraints / Description
id	BigInteger	PK	Auto-incrementing primary key
case_id	BigInteger	FK (cases_case)	Parent Case folder link
title	VarChar(255)	-	Friendly title/label for the file
file	VarChar(100)	-	Storage path to the uploaded document
file_type	VarChar(20)	-	Choices: pdf, txt, docx, or other (Default)
extracted_text	Text	-	Raw text content extracted from file contents
uploaded_at	DateTime	-	Automatically set on upload

Evidence Files

Column Name	Data Type	Key Type	Constraints / Description
id	BigInteger	PK	Auto-incrementing primary key
evidence_file_id	BigInteger	FK (evidence_evidencefile)	Source document link
chunk_index	Integer	-	Sequence number of chunk within file
content	Text	-	Snippet text block
qdrant_point_id	VarChar(100)	Unique	UUID matching vector mappings inside vector storage database
created_at	DateTime	-	Automatically set on generation

Evidence Chunks

Form Structure

The image shows a dark-themed login interface. On the left, a rounded rectangle contains the text "WELCOME BACK!" in large, bold, light blue letters, followed by "Login to access your secure panel." in smaller white text. On the right, another rounded rectangle contains the title "Secure Login" in white. Below the title are two input fields: "Username" and "Password", each with a light blue underline. A light blue button with the text "INITIALIZE SIGN-IN" in white is positioned below the password field. At the bottom of the right section, the text "New terminal user?" is followed by a light blue link "Register Here". The entire interface is set against a dark blue background with a subtle grid pattern.

WELCOME BACK!
Login to access your secure panel.

Secure Login

Username

Password

INITIALIZE SIGN-IN

New terminal user? [Register Here](#)

Form Structure

Create Terminal Account

Username

Email Address

Password

Confirm Password

REGISTER ACCESS

Existing credential? [Login Here](#)

JOIN US!

Create an account to start your journey.

Subscription Page

Subscription Page

1 Month

6 Months

1 Year

SILVER

₹149.00

Basic access to the Digital Forensics platform for 30 days. Includes forensic dashboard access, AI investigation support, and secure evidence management.(ollama)

1 Month Access

Subscribe

GOLD

POPULAR

₹199.00

access to the Digital Forensics platform. Ideal for long-term investigations with uninterrupted AI-powered forensic analysis with GLM

1 Month Access

Subscribe

PREMIUM

₹299.00

premium access to the Digital Forensics system. Best for continuous forensic investigations with long-term secure storage and advanced AI forensic assistance WITH OLLAMA AND GLM

1 Month Access

Subscribe

CONCLUSION

Advantages, Future Scope & Conclusion

Future Enhancements

- Image and video forensic analysis
- Real-time monitoring and alerts
- Predictive threat intelligence
- Multi-language support
- Timeline Reconstruction

- ✓ The system bridges the gap between raw digital evidence and actionable forensic intelligence.

→ **Faster investigations** — AI reduces manual effort significantly

→ **Better context** — semantic understanding over keyword matching

→ **Scalable architecture** — handles growing evidence volumes

Conclusion

This project demonstrates how **RAG and vector databases** can transform digital forensics — enabling intelligent, context-aware, and scalable investigation workflows that outperform traditional methods.

Thank You

Questions & Discussion Welcome

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